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Enhancing Communicative Skills For Combined Operations Through A Multimodal XR-AI Training System

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The views contained in this article are the authors' alone and do not represent the views of KMA.

Abstract: Communicative interoperability is vital in multinational coalition warfare. While current Extended Reality (XR) systems excel in procedural motor training, this study proposes developing the KMA-XR-AI system to expand this scope to the communicative 'fog of war.' The system is intended to synergise multimodal sensing with adaptive AI to complement traditional XR capabilities. By fusing bio-physiological and acoustic signals via a Hybrid Sensing Module, the system would calculate a real-time Cognitive Load Index (CLI) to drive a Smart Scaffolding Protocol. This intervention will be designed to distinguish between linguistic deficits and stress-induced inhibition during tactical verbal interactions, and ultimately, optimise verbal proficiencies essential for mission success, offering a technological evolution for future military training.

Bottom-line-up-front: The KMA-XR-AI system demonstrates how integrating advanced technology can transform Military English into a resilient combat skill, ensuring officer cadets maintain effective communication and command integrity during high-stress multinational

operations.

Problem statement: How can military academies evolve their training to help officer cadets overcome stress-induced communication failures in multinational operations?

So what?: Military educators must recognise stress-induced communication failures as a tactical risk and evolve training paradigms accordingly. By strategically leveraging high-impact tools, they can redefine Military English as a resilient combat skill, transforming a potential vulnerability into a decisive operational advantage for multinational missions.

KMA's Multi-User Extended Reality (XR) Environment

To overcome the limitations of live-fire exercises, specifically budgetary, temporal, and spatial constraints, the Korea Military Academy (KMA) has proactively integrated advanced Extended Reality (XR) tactical simulators. This XR training system provides high-fidelity environments for individual marksmanship and squad-level manoeuvres. Built on the Unity engine, the environment utilises advanced physics plugins and a dedicated ballistics module that calculates trajectory shifts based on wind direction, humidity, and gravity drop. This ensures that kinetic engagements adhere to real-world physics, requiring officer cadets to adjust their aim points dynamically.

The training facility consists of 10 networked individual simulator units. Each unit features three large flat projection screens arranged in an open-pentagonal configuration. The screens surround the user to provide a panoramic field of view covering the front and flanks, while the rear section remains open to serve as an entrance. This design immerses participants in the visual environment without the isolation of a closed dome. During the exercise, participants utilise high-fidelity K2 training rifles, which are identical in weight and appearance to the standard-issue Republic of Korea (ROK) Army service rifle, while also replicating the kinetic recoil of live fire. Additionally, they wear haptic feedback suits that deliver localised vibration cues upon virtual impact. This ensures that the physical cognitive load of active marksmanship and the physiological stress of sustaining virtual injuries compete with the cognitive resources necessary for tactical performance.

Communicative Interoperability in Combined Operations

While these systems excel at physical simulation, the strategic reality of combined operations and multinational operations necessitates a focus on communicative interoperability. The failure to coordinate through a common operational language can lead to mission paralysis and, in worst-case scenarios, fratricide. Historical records are replete with such communicative failures, with a particularly illustrative case occurring during the Spanish Civil War in 1937.

The Irish Brigade, which primarily used English, was mistakenly engaged by a Spanish-speaking Falangist unit from the Canary Islands. Lacking a shared proficiency in Spanish or any established common operational language to identify themselves, the two allied units exchanged fire for over an hour, resulting in avoidable deaths due to a complete linguistic disconnect.[1]

Most recently, this danger has re-emerged in modern multinational operations. In 2024, intelligence reports indicated that North Korean troops deployed to the Kursk region engaged in fratricide with allied Russian forces.[2] The incident was attributed to a severe shortage of interpreters and the North Korean soldiers' inability to comprehend Russian tactical commands or IFF (Identification Friend or Foe) challenges, proving once again that communicative interoperability is a survival requirement. These incidents underscore that without a proficient command of the operational language, the ‘fog of war’ becomes exponentially denser, turning allies into unintentional adversaries.

Advancing the Scope of Current Training Paradigms

The lethal consequences of communicative failure, as seen in the abovementioned multinational tragedies, demonstrate that communicative interoperability is not a secondary skill but a primary component of survival. To prevent fratricide and ensure mission success, KMA officer cadets must be able to maintain precise communicative coordination in English —the essential operational language for ROK-U.S. combined operations—even under the visceral stress and acoustic chaos of the battlefield. However, for KMA officer cadets, there is a significant gap between this strategic necessity and the current state of training.

While frequent operational exchanges with U.S. forces are ideal, logistical constraints largely confine training to Korean-led classroom instruction. To bridge this gap, KMA incorporates guest lectures and social interactions with U.S. officers; however, these remain limited to academic or social exchanges, lacking the tactical intensity of live combat. Consequently, traditional environments fail to replicate the physiological pressure known to degrade verbal performance—an impact that is disproportionately severe for second language (L2) proficiency.[3] This suggests that an officer cadet who excels in a classroom environment may still experience a communicative breakdown amid the chaos of a high-pressure combined operation.

To address this, KMA’s XR training system offers a powerful platform; yet there is a distinct opportunity to further enhance its operational utility. As Shin et al. observe, while research on digital twin-based combat platforms is robust, it remains heavily focused on the physical fidelity of the battlefield.[4] This creates a critical training asymmetry: officer cadets achieve

kinetic proficiency through advanced simulators, yet the system’s very design—which isolates squad members to maximise individual immersion—creates a 'tactical vacuum' for communication. The physical barriers preclude essential nonverbal cues such as eye contact and hand signals, while the open-rear environment is acoustically suboptimal for verbal coordination.

The Path Toward ‘Training 2.0’

To fill this ‘tactical vacuum’ and address the critical need for enhancing communicative competence, this research embodies the 'Training 2.0' vision—the central theme of TMAF 2026—by moving beyond static simulation toward an adaptive, communicative ecosystem. This approach expands the scope of training from conventional physical and tactical proficiency to include communicative readiness as a core combat capability.

The Efficacy of XR in Educational Contexts

Recent scholarship has largely reached a consensus on the pedagogical benefits of XR, positioning it as a transformative medium in education. A comprehensive umbrella review by Dong et al., which synthesised findings from 20 meta-analyses, concluded that XR interventions consistently yield a medium-to-large effect size (Cohen’s d = 0.723–0.951) compared to traditional instruction.[5] This significant empirical evidence suggests that immersive environments are not merely technological novelties but potent tools that substantially enhance knowledge retention and skill acquisition. Beyond improvements in academic performance, XR has been proven to be highly effective in boosting learner motivation and engagement by providing vivid learning resources and interactive environments that capture students’ attention.[6] By visualising instructional content and allowing learners to directly experience virtual environments, XR bridges the gap between theoretical knowledge and practical application, enabling students to understand complex subjects more intuitively.[7] Furthermore, research indicates that XR fosters a learner-centred environment where users can actively explore and control their learning process, which leads to increased self-efficacy and enjoyment with respect to the subject matter.[8] Ultimately, the integration of XR into educational contexts shifts the learning paradigm from passive observation to active experiential learning, validating its efficacy as a robust and essential educational instrument.[9]

From Kinetic Precision to Communicative Synergy

The application of XR in the military domain has demonstrated remarkable success, particularly in establishing a robust foundation for psychomotor training. Current research has primarily excelled in enhancing kinetic proficiency, with a sophisticated focus on marksmanship and weapon handling.[10] For instance, innovations, such as Wei et al.'s haptic feedback systems, have significantly advanced the field by providing realistic physical mechanics of engagement.[11] Building on these achievements, there is a strategic opportunity to expand this high-fidelity immersion into the realm of communicative interoperability.

Recognising that tactical competence in combined operations is a synergy of both physical precision and communication competence, the next logical evolution for XR is the integration of communicative dimensions. Since the inability to coordinate fire missions can be as critical as a missed shot, evolving XR from "silent" simulations into "communicative" ecosystems is essential. This transition involves leveraging high-fidelity audio hardware to facilitate realistic English language interaction, ensuring that the XR space reflects the full complexity of the battlefield.

Furthermore, extant psychophysiological research provides a proven baseline for this expansion. While pioneers such as Hyun et al. and Ku et al. have successfully utilised XR to measure physiological stress during shooting, their work paves the way for a broader investigation.[12] Building upon these foundations, this study aims to expand the functional scope of XR beyond kinetic precision and physiological monitoring to encompass the critical dimension of communicative competence. This evolution enables a more holistic system capable of monitoring and adapting to a learner's stress levels in real time, bridging the gap between physical action and linguistic resilience.

Distinctive Motivation and Affective Barriers in Military English Learning

As XR environments evolve to support high-level communicative synergy, the pedagogical approach must account for the unique psychological profile of non-native officer cadets. A critical distinction must be made between developing general communicative competence and mastering English as L2 within a combined operations context. According to Kim et al., proficiency or motivation in general English does not automatically translate into a commitment to military-specific English.[13] For Korean officer cadets, this learning process is not merely an academic exercise but a survival-based professional requirement for ROK-U.S. combined operations, demanding a distinct motivational framework centred on operational utility.

However, even with high instrumental motivation, the necessity of communicating in L2 introduces a formidable psychological barrier: Foreign Language Anxiety (FLA).[14] Within the high-pressure environment of combined operations, FLA acts as a decisive emotional variable that can effectively decouple an officer cadet's tactical expertise from their communicative performance. Under the 'Affective Filter Hypothesis,'[15] intense anxiety functions as a mental blockade, particularly for L2 learners, hindering the dynamic cycle of communication by impairing an officer cadet's ability to process incoming reports and issue clear, outgoing directives in Military English.

Consequently, for these future officers, the fear of committing a tactical error due to a linguistic mistake can lead to communicative breakdown. This underscores a vital reality: no matter how profound an officer cadet’s tactical knowledge may be, it cannot be translated into operational action if high levels of anxiety prevent them from communicating effectively. In a tactical environment, the stakes of such failure are exceptionally high, as a linguistic mistake is not merely a personal error but a direct threat to the lives of fellow soldiers and the overall success of the unit’s mission. Ultimately, in the context of ROK-U.S. combined operations, English communicative proficiency is not merely an auxiliary skill but a mission-critical tactical requirement that directly determines an officer's operational reliability.

Current Study

Aligning with the 'Training 2.0' vision and the evolutionary trajectory of military XR, this study proposes the KMA-XR-AI system—a novel framework designed to operationalise communicative readiness in high-stakes environments. Moving beyond the 'kinetic-only' focus of traditional drills, this research explores how augmenting existing XR systems can transform the training space into a responsive laboratory for mission success. The primary objective is to investigate the architectural and functional requirements of the KMA-XR-AI system to identify linguistic errors and monitor learner anxiety in real-time. By providing immediate support based on these observations, an officer cadet's tactical wisdom could be reliably translated into operational action, rather than being hindered by the psychological strain of foreign language communication. To guide this investigation, the current study addresses the following research questions:

- RQ1 (Hardware Requirements): What are the core hardware-based architectural requirements to effectively capture multimodal learner data—including vocal features and physiological signals—in real-time during tactical engagements?
- RQ2 (Interactive Support): What software-driven interaction logics and scaffolding mechanisms are necessary to identify linguistic errors and mitigate psychological

overload to maintain continuous dialogue flow in English?

- RQ3 (Pedagogical Efficacy): To what extent does the KMA-XR-AI system reduce Foreign Language Anxiety (FLA) and increase Willingness to Communicate (WTC), thereby enhancing actual Military English communicative skills and tactical operational performance?

The Proposed "Training 2.0" System

To realise the strategic shift from kinetic-focused training to a more holistic, communicative-centric paradigm, this study proposes the KMA-XR-AI system. This integrated ecosystem is designed as a bio-adaptive architecture that transforms the current XR environment into a responsive training space. As these combined technologies form the crux of the proposed framework, the subsequent sections elaborate on how the ideas and design concepts for the two main components—a Hybrid Acoustic-Bio Sensing Module and the AI Tactical Communicator—were conceptualised. Specifically, the following sections explicitly outline the background, operational inspirations, and thought processes that went into developing the hardware and the internal workings of the AI software. By symbiotic integration of specialised hardware and adaptive software, the proposed framework aims to address the systemic and functional requirements identified in RQ1 and RQ2.

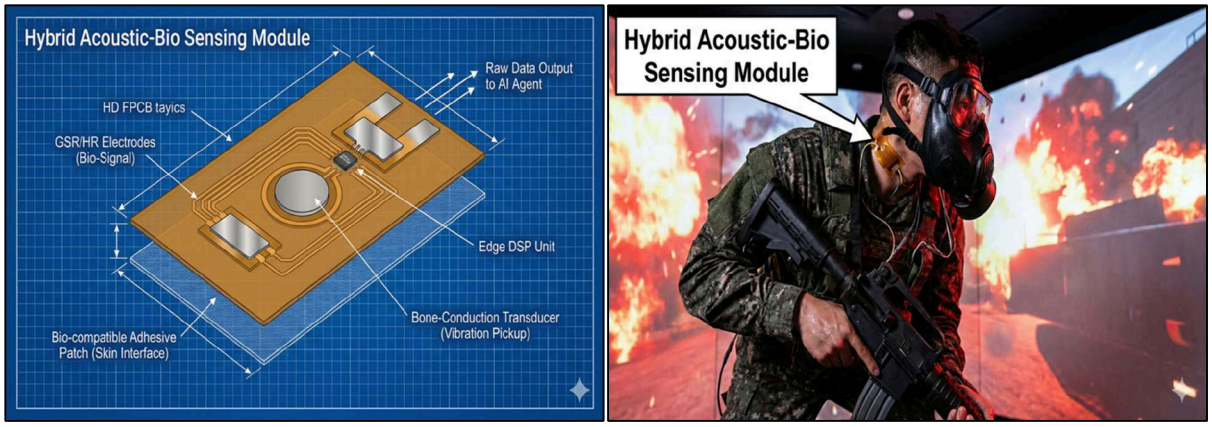
Hybrid Acoustic-Bio Sensing Module

In response to RQ1, which seeks the core hardware requirements for real-time multimodal data capture, we propose developing the Hybrid Acoustic-Bio Sensing Module (HSBSM). While KMA's current XR infrastructure is operational for tactical drills, conventional ear-hook microphones frequently fail to function reliably in high-noise military environments or during intense physical exertion. To overcome these limitations, this study proposes a proprietary wearable device specifically designed to operate in extreme conditions, including CBRN scenarios where gas masks are mandatory. This module serves as the primary hardware interface for the proposed framework, ensuring data integrity through the following specialised design and processing strategies.

Design of Hardware Form Factor Using Flexible PCB

To address the persistent issue of sensor displacement inside a gas mask, we propose an ergonomic housing design utilising Flexible PCB (FPCB) technology. Unlike rigid sensors, this flexible form factor is intended to allow the microphone and bio-sensors to maintain consistent skin contact without causing facial pressure or discomfort during intense tactical

manoeuvring. The FPCB-based module is engineered with bio-compatible adhesive properties, ensuring that the sensor remains conformal to the skin's curvature even during rapid, high-frequency motions or excessive perspiration.



Conceptual Design and In-situ Application of the Hybrid Acoustic-Bio Sensing Module; Source: Gemini, Google.

This design aims to ensure data integrity even when the user is physically active or sweating. To achieve this, the FPCB substrate integrates a series of micro-electrodes that maintain direct contact with the skin, functioning as a multi-channel sensing interface for both electrocardiological and electrodermal activities.

Proposed Acoustic Signal Processing Pipeline

Acknowledging that battlefield noise levels frequently exceed 100dB, rendering standard microphones ineffective, this research plans to implement a three-stage signal processing pipeline to ensure clear communication:

Stage	Core Technology	Primary Function / Role
Stage 1	Bone-Conduction Technology	Mechanically isolates airborne noise by capturing vocal cord vibrations directly from the user.
Stage 2	DSP & ANC (Beamforming)	Eliminates residual noise and restores acoustic fidelity through digital refinement and active cancellation.
Stage 3	Feature Extraction & Analysis	Quantifies paralinguistic markers (e.g., Pitch, Jitter, Pause) to calculate the real-time Cognitive Load Index (CLI).

Three-Stage Signal Processing Pipeline of the Hybrid Acoustic-Bio Sensing Module; Source: Author.

- *Stage 1 (Bone-Conduction)*. The system will utilise bone-conduction technology, employing a sensor designed to capture vocal cord vibrations directly from the neck or mastoid area. This approach aims to mechanically bypass airborne acoustic interference.
- *Stage 2 (DSP & ANC).* The second stage is designed to leverage a Digital Signal Processing (DSP) unit. We will apply beamforming algorithms to spatially isolate the

speaker's voice from ambient explosions, while simultaneously utilising Active Noise Cancellation (ANC) to further purify the signal before it is processed by the AI.

- *Stage 3 (Feature Extraction & Analysis).* In the final stage, the purified signal is processed to extract key paralinguistic markers, including pitch, jitter, and pause patterns. These features serve as the primary indicators for calculating the real-time Cognitive Load Index (CLI), allowing the system to detect physiological stress and potential linguistic breakdown.

Strategy for Bio-Signal Fusion

Concurrently, the module is engineered to capture raw physiological signals through its integrated electrodes at a high-resolution sampling rate (100Hz). The fusion strategy focuses on two primary modalities:

- *Heart Rate Variability (HRV).* The system samples raw electrocardiogram (ECG) signals to identify R-R intervals (i.e., the time between consecutive heartbeats). By analysing the variance in these intervals, the system quantifies autonomic nervous system stress.
- *Galvanic Skin Response (GSR).* The module measures Skin Conductance (SC) by detecting minute changes in electrical resistance caused by sweat gland activity. This provides a direct correlate for emotional arousal and 'panic' states during combat simulations.

AI Tactical Communicator

In response to RQ2, we propose the AI Tactical Communicator, the software-driven core of the KMA-XR-AI system. This agent is envisioned to function as a Virtual U.S. Army Team Leader, delivering direct tactical commands and real-time situational updates in English. Unlike conventional chatbots, it operates as a dual-purpose agent: a mission-driven leader and an adaptive language instructor. Its primary intelligence lies in diagnosing the nature of communicative breakdowns—determining whether they stem from a linguistic deficit (lack of L2 knowledge) or stress-induced cognitive overload. By utilising this real-time diagnosis, the AI functions as a pedagogical scaffold, adjusting its language complexity and providing tailored prompts to ensure that the officer cadet maintains dialogue flow and successfully completes the tactical mission. The specific mechanisms for this adaptive instruction are detailed in the following subsections.

Proposed Modular Scenario Structure

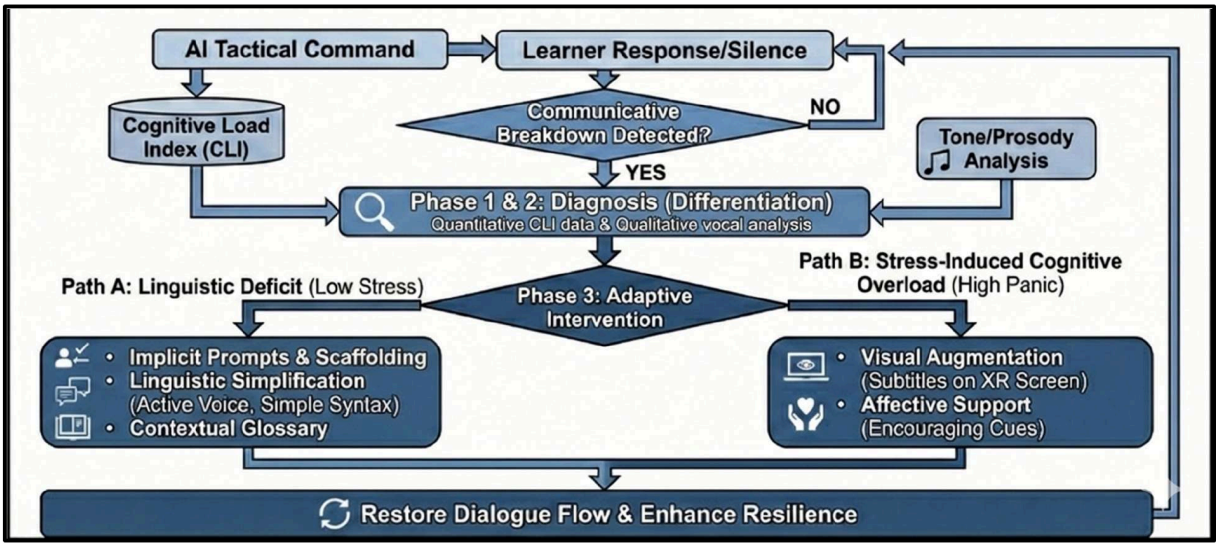
The AI's decision-making logic follows a Context-Utterance-Action (CUA) modular structure. It first establishes the Context by calculating the CLI, analyses the Utterance through prosodic validation, and finally determines the most effective scaffolding Action.

Strategy for Cognitive Load Index (CLI) Modelling

The system aims to establish a real-time Cognitive Load Index (CLI) by analysing the correlation between acoustic features (i.e., Pitch, Jitter, Pauses) and bio-signals (i.e., HRV and GSR) using a regression analysis model. This index will serve as the baseline for distinguishing stress-induced errors from linguistic errors.

Proposed Smart Scaffolding Protocol

The AI agent operates on a structured three-phase intervention protocol to restore tactical communication flow:



AI Tactical Communicator Decision Logic & Smart Scaffolding Protocol; Source: Gemini, Google.

Phase 1: Detection & Quantitative Differentiation. The system continuously monitors the dialogue for a "communicative breakdown"—defined as prolonged silence, syntactic collapse, or failure to respond to tactical commands in English. Upon detection, the CLI Modelling Engine performs an immediate quantitative assessment. By processing synchronised bio-signals and acoustic data in real-time through regression analysis, the engine identifies the root cause of the breakdown. It classifies the state into either a Linguistic Deficit (i.e., low physiological stress, indicating a struggle with English formulation) or Stress-Induced Cognitive Overload (i.e., high physiological arousal, indicating a psychological blockade).

Phase 2: Qualitative Prosodic Validation. To prevent "false positives"—such as misinterpreting high heart rates from physical exertion or intentional tactical silence as psychological panic—the system enters a qualitative validation stage. A dedicated Tone/Prosody Analysis module examines the learner's vocal markers, specifically pitch fluctuations, jitter (frequency instability), and speech rate. This phase functions as a sophisticated filter to distinguish between:

- Intentional Tactical Engagement: Situations where an officer cadet is silent due to intense task focus or shouting with stable, controlled urgency.
- Unintentional Communicative Collapse: Situations where the cadet experiences vocal jitter, pitch spikes, or dysfunctional stuttering caused by genuine cognitive paralysis.

Only when these prosodic markers align with the high CLI data is the diagnosis confirmed, ensuring the AI does not interrupt an officer cadet who is simply in a state of high tactical concentration.

Phase 3: Adaptive Scaffolding Intervention. Based on the validated diagnosis, the AI activates one of two specialised "Paths" to restore the dialogue flow:

- Path A for Linguistic Scaffolding (for Linguistic Deficits): The AI reduces the linguistic barrier to entry without altering tactical substance. The agent applies Linguistic Modification by simplifying syntax (e.g., converting passive commands to active voice) and providing Implicit Prompts—guiding the officer cadet toward the correct command rather than providing intrusive, flow-breaking corrections. A Contextual Glossary for non-tactical vocabulary is also projected on-screen to assist word retrieval.
- Path B for Multimodal Support (for Cognitive Overload): To mitigate the effects of the 'Affective Filter,' the AI provides Visual Augmentation by displaying critical mission data or subtitles directly on the XR interface to compensate for auditory processing failure. Simultaneously, it delivers Affective Support through calm, encouraging cues designed to lower anxiety and restore the officer cadet's sense of self-efficacy in a chaotic environment.

Methodology for An Experiment

To address RQ3, this study proposes to employ a Within-Subjects Counterbalanced Design (3x3 Latin Square). This approach is specifically chosen to neutralise individual variability in L2 proficiency and tactical experience by allowing each cadet to serve as their own control.

Furthermore, this design ensures ethical parity by providing all participants with equal access to every training condition while maximising statistical power through the systematic rotation of scenarios and interventions.

Participants

The study will recruit 60 officer cadets from KMA. Although the sample size is smaller than that of typical between-subjects studies, the within-subjects design provides equivalent statistical power because every participant generates data for all three experimental conditions. Prior to the experiment, participants will be screened for baseline English proficiency to ensure homogeneity across the sample.

Instrumentation

To rigorously validate the system's impact, this research employs a triangulation approach that correlates subjective psychological metrics, objective physiological data, and tactical performance outcomes:

Subjective Metrics (Psychological Shifts)

Validated scales adapted for the military context—specifically FLA, WTC, and Self-Efficacy —will be administered. Officer cadets will complete these scales immediately before and after the experimental sessions to track longitudinal shifts in their psychological state and communicative confidence.

Objective Metrics (Physiological Stress)

For the XR experimental groups, real-time stress levels will be quantified using the Hybrid Sensing Module. These acoustic and biological data points are integrated into the CLI via the regression analysis model, providing a direct, empirical link between the hardware's sensing capabilities and the officer cadet's internal state.

Performance Metrics (Tactical Proficiency)

Task performance will be evaluated by experienced tactical officers using a rubric based on the U.S. Army's "Warrior Leader Skills." This assessment focuses on the accuracy of military terminology, adherence to radio protocols, and overall mission success.

Three Distinct Combat Modules

To prevent "learning effects", where carryover knowledge from one session influences the next, three distinct, non-overlapping combat scenarios of equivalent difficulty will be selected from the U.S. Army's Warrior Leader Skills doctrine. For example, Scenario A (Call for Fire),

Scenario B (MEDEVAC), and Scenario C (SALUTE Report) constitute the core tasks.[16] Critically, the tactical and linguistic equivalence of these three modules will be pre-validated through expert review by three senior tactical officers to ensure that performance variance is not a function of task difficulty.

Experimental Design (Latin Square)

Participants will be randomly divided into three groups (n = 20 in each group). Each group rotates through three conditions across the three scenarios. Condition 1 (XR + AI) is the experimental condition; Condition 2 (XR Only) measures baseline immersion; and Condition 3 (Classroom) serves as the control.

Group	Session 1 (Scenario A: Call for Fire)	Session 2 (Scenario B: MEDEVAC)	Session 3 (Scenario C: SALUTE)
Group 1	XR + AI (Experimental)	XR Only (Baseline)	Classroom (Control)
Group 2	Classroom (Control)	XR + AI (Experimental)	XR Only (Baseline)
Group 3	XR Only (Baseline)	Classroom (Control)	XR + AI (Experimental)

Experimental Design (3 x 3 Latin Square); Source: Author.

Procedure

The procedure will begin with a pre-test to assess baseline English proficiency and anxiety levels. The core intervention comprises three weeks of rotation sessions, during which groups perform their assigned conditions for Scenarios A, B, and C. To mitigate potential carryover effects, a strict one-week “wash-out period” is maintained between each session. Immediately following each 15-minute task, participants complete post-task measures of anxiety and performance, while the system logs physiological data for CLI calculation. The study concludes with a post-interview to gather qualitative feedback regarding the perceived helpfulness of the AI agent compared to traditional and XR-only methods.

Data Analysis

Data analysis will be conducted using STATA, employing a one-way repeated measures ANOVA as the primary method. To ensure statistical rigour, the order of participation is included as a covariate to control for potential sequence effects. The analysis tests for significant main effects of the training condition on communicative performance, FLA, WTC, and the CLI. If a significant main effect is observed, Bonferroni-adjusted pairwise comparisons will be performed to isolate specific causal factors. These comparisons will specifically evaluate the "pure AI effect" (XR+AI vs XR Only), the "immersion effect" (XR Only vs Classroom), and the overall system efficacy (XR+AI vs Classroom) compared to current education standards for Military English learning.

Implications and Conclusion

Academic and Practical Implications

This research pioneers a communicative-centric paradigm by integrating real-time bio-signal analysis with L2 Military English learning within a high-fidelity XR environment. By effectively addressing the "kinetic bias" prevalent in current military training, this study will provide empirical evidence for the "Training 2.0" paradigm. By scientifically quantifying the CLI, this research will demonstrate that communicative interoperability under pressure is not merely a soft skill, but a measurable and trainable tactical asset. Maintaining clear communication under duress is as fundamental to the success of combined operations as the traditional pillars of marksmanship and manoeuvre.

Technologically, the development of the "Hybrid Acoustic-Bio Sending Module" represents a significant engineering breakthrough. By combining ergonomic Flexible PCB (FPCB) technology with a dual-stage processing pipeline that includes bone-conduction and DSP + ANC, this device could help address the persistent challenge of capturing clear voice data in gas masks and high-noise environments exceeding 100dB. Moreover, this innovation would remove the primary hardware barrier that previously hindered the adoption of speech-based AI in tactical training, ensuring data integrity even during intense physical manoeuvring.

Complementing this hardware is the pedagogical innovation of the AI Tactical Communicator. Unlike static scripting, this agent will utilise a three-phase Smart Scaffolding Protocol to distinguish whether a communication breakdown stems from a "Linguistic Deficit" or "Stress-Induced Cognitive Overload." By implementing Linguistic Modification strategies based on Abedi et al.—such as syntactic simplification and contextual glossaries—the system establishes a new standard for Intelligent Tutoring Systems in high-stakes domains, specifically for military training in combined operations.

Broader Impact and Scalability

Beyond its immediate military applications, the proposed system offers far-reaching implications for other high-stakes professions. The AI agent’s "Context-Utterance-Action" modular structure enables seamless adaptation to sectors such as high-risk counterterrorism operations and fire-rescue command systems. Furthermore, the core mechanism of managing communicative breakdown under stress can also be expanded to first language (L1) communication training for emergency responders in chaotic disaster zones.

Significantly, this framework holds immense potential for global military interoperability. While developed for ROK officer cadets, the system is directly applicable to other non-English-speaking allies who conduct combined operations with the U.S. and the United

Kingdom. This is particularly relevant for most European NATO member states, where standardising Military English is essential for collective defence. By mitigating the "Affective Filter" during multinational exercises, this technology ensures that linguistic barriers do not compromise mission success. Ultimately, such scalability demonstrates how military-oriented XR technology can be repurposed to enhance both international security and public safety by ensuring clear communication channels under extreme psychological pressure.

Conclusion

The proposed 'KMA-XR-AI System' is designed to advance military training by further integrating Military English as a core tactical asset for combined operations. By synergising hybrid hardware sensing with adaptive AI scaffolding, this research aims to strengthen the communicative skills of officer cadets within complex multinational environments. However, a primary limitation of the current study is its conceptual nature; as the proposed technological framework and experimental design have yet to be empirically executed, concrete data regarding its practical efficacy is not currently available. Accordingly, the next phase of this research will focus on the empirical execution of the triangulated experimental design detailed herein. By putting this established framework into practice, our subsequent studies aim to rigorously verify its pedagogical efficacy through objective performance, physiological, and acoustic data. Ultimately, this study seeks to enhance the operational interoperability of combined forces, ensuring that Military English serves as a decisive bridge for mission success in high-stakes global environments.

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[16] These scenarios represent standardized military communication protocols. Specifically, a Call for Fire is a procedural request for indirect fire support onto a target; MEDEVAC (Medical Evacuation) is a formalized process for extracting casualties from the battlefield, typically using a standard 9-line format; and a SALUTE report is a mnemonic framework (Size, Activity, Location, Unit, Time, and Equipment) used to relay intelligence regarding enemy forces.